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Stomata are found mainly on the underside of leaves of dicotyledonous plants. A student painted a layer of nail varnish on the lower epidermis of a leaf. She peeled off the dry layer of nail varnish and placed it on a microscope slide. The student was able to see the impressions of the stomata on the varnish using an optical microscope. She then determined the mean diameter of the stomata.

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Describe how the student could use an eyepiece graticule to determine the mean diameter of stomata.

[3 marks]

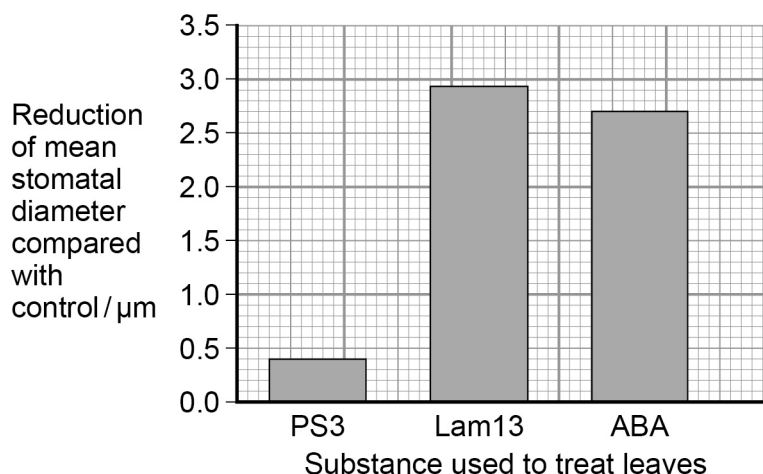
[Extra space]



A scientist investigated the effect of treating the leaves of one species of plant with three different substances. These substances **reduce** the stomatal diameter. He compared the mean diameter of stomata after treating the leaves with these substances with the mean stomatal diameter on control leaves treated with distilled water.

The scientist's results are shown in **Figure 6**.
The mean stomatal diameter of the control leaves was 7.5 μm .

Figure 6



0 6 . 2

Calculate the ratio of mean stomatal diameter of leaves treated with PS3 to those treated with ABA.

[2 marks]

Answer = _____ : 1

Question 6 continues on the next page

Turn over ►



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ABA is a substance that some plant species produce when little water is available.

Explain why producing ABA may help these species survive in dry conditions.

[2 marks]

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Many species of plants can be infected by powdery mildew which is spread by microscopic spores in the air.

Suggest how treatment with Lam13 might protect plants against powdery mildew infection.

[1 mark]

8

